

$$AM = \frac{a+b}{2}, \quad GM = \sqrt{ab}, \quad H = \frac{2ab}{a+b}$$

Chapter-5 Sequence and Series

• Sequence

- A sequence is a function defined on set of positive integers.

i.e. $f: \mathbb{N} \rightarrow \mathbb{R}$

e.g. 2, 4, 6, --- ($f: \mathbb{N} \rightarrow \mathbb{R}$ defined by $f(n) = 2n$)

Types of sequence

① Arithmetic sequence or progression

- A sequence is called arithmetic if difference between two successive terms is always same.

e.g. 2, 4, 6, 8, ---

* Arithmetic mean (A.M.)

→ The middle term or terms between first and last term of arithmetic sequence is called Arithmetic mean.

If a, A, b are in A.P then,

$$A - a = b - A$$

$$\text{or, } 2A = a + b$$

$$\text{or, } A = \frac{a+b}{2} \Rightarrow \therefore \text{A.M} = \frac{a+b}{2}$$

If there are n arithmetic means then

$$d = \frac{b-a}{n+1}$$

$$\text{So, } t_1 = a$$

$$t_2 = a + d$$

$$t_3 = a + 2d \text{ \& so on.}$$

② Geometric sequence

→ A sequence is called Geometric if ratio between two successive terms is always same.

e.g. 2, 4, 8, 16, ---

* Geometric mean (G.M)

→ The middle term or terms between first and last term of Geometric sequence is called geometric mean.

→ It is denoted by G.M

→ If a, G, b are in G.P then

$$\frac{G}{a} = \frac{b}{G}$$

$$\text{or, } G^2 = ab$$

$$\text{or, } G = \sqrt{ab} \Rightarrow \therefore \text{G.M} = \sqrt{ab}$$

If there are n geometric means then,

$$r = \left(\frac{b}{a} \right)^{\frac{1}{n+1}}$$

$$\text{then, } t_1 = a$$

$$t_2 = ar$$

$$t_3 = ar^2$$

③ Harmonic sequence

→ A sequence is called harmonic if reciprocal of each term form Arithmetic sequence.

e.g. $\frac{1}{2}, \frac{1}{4}, \frac{1}{6}, \frac{1}{8}, \dots$

* Harmonic mean

→ The middle term or terms between first and

last term of harmonic sequence is called harmonic mean.

- It is denoted by H.M.

- If a, H, b are in H.P

$$\Rightarrow \frac{1}{a}, \frac{1}{H}, \frac{1}{b} \text{ " " A.P}$$

$$\Rightarrow \frac{1}{H} - \frac{1}{a} = \frac{1}{b} - \frac{1}{H}$$

$$\Rightarrow \frac{2}{H} = \frac{1}{a} + \frac{1}{b}$$

$$\Rightarrow \frac{2}{H} = \frac{a+b}{ab}$$

$$\Rightarrow H = \frac{2ab}{a+b}$$

Note

① Any three consecutive terms of A.P are $a-d, a, a+d$

② Any three consecutive terms of G.P are $\frac{a}{r}, a, ar$

Theorem:

- If A.M, G.M and H.M are arithmetic mean, Geometric mean and Harmonic mean respectively between two unequal positive numbers a and b then.

$$i) G.M^2 = A.M \times H.M$$

$$ii) A.M > G.M > H.M$$

Proof.

i) We have,

$$A.M = \frac{a+b}{2}, \quad G.M = \sqrt{ab}, \quad H.M = \frac{2ab}{a+b}$$

$$R.H.S = A.M \times H.M = \frac{(a+b)}{2} \times \frac{2ab}{(a+b)}$$

$$= ab$$

$$= (\sqrt{ab})^2$$

$$= G.M^2 = L.H.S$$

proved
z.ii) $A.M > G.M > H.M$

$$A.M - G.M = \frac{a+b}{2} - \sqrt{ab}$$

$$= \frac{a+b-2\sqrt{ab}}{2}$$

$$= \frac{\sqrt{a}^2 - 2\sqrt{a}\sqrt{b} + \sqrt{b}^2}{2}$$

$$A.M - G.M = \frac{(\sqrt{a} - \sqrt{b})^2}{2} > 0$$

$$\text{or, } A.M - G.M > 0$$

$$A.M > G.M \rightarrow \text{(i)}$$

Also, we have,

$$G.M^2 = A.M \times H.M$$

$$\text{or, } \frac{A.M}{G.M} = \frac{G.M}{H.M}$$

$$G.M > H.M$$

since from (i) we can conclude,

$$G.M > H.M \rightarrow \text{(ii)}$$

combining (i) & (ii)

$$A.M > G.M > H.M \quad \text{✓}$$

$$GM^2 = AM \times H.M$$

• Exercise 5.1

1.

a) Find two numbers whose arithmetic mean is 20 and geometric mean is 16.

→ Soln,

Let A.M & G.M are arithmetic ~~mean~~ ~~and~~ mean and Geometric mean of two number x and y .

$$A.M = 20$$

$$\text{or } \frac{x+y}{2} = 20$$

$$\text{or } x+y = 40$$

$$\text{or } x = 40 - y \rightarrow (i)$$

$$G.M = 16$$

$$\text{or } \sqrt{xy} = 16$$

∴ Squaring on both sides,

$$xy = 256$$

$$\text{or } x = \frac{256}{y} \rightarrow (ii)$$

Putting the value of x , we get

$$40 - y = \frac{256}{y}$$

$$\text{or } 40y - y^2 = 256$$

$$\text{or } y^2 - 40y + 256 = 0$$

$$\text{or } y^2 - 8y - 32y + 256 = 0$$

$$\text{or } y(y-8) - 32(y-8) = 0$$

$$\text{or, } y(y-8)(y-32)=0$$

Either,

$$y = 8, 32$$

When $y=8$

$$x = \frac{256}{8} = 32$$

When $y=32$

$$x = \frac{256}{32} = 8$$

$\therefore$ Two numbers are 8 and 32 or 32 and 8.

b) The A.M between two numbers is 40 and their H.M is 30. Find the numbers.

→ Soln,

Let A.M and H.M be Arithmetic mean and Harmonic mean of two numbers a and b .

$$\text{A.M} = 40$$

$$\text{or, } \frac{a+b}{2} = 40$$

$$\text{or, } a+b=80$$

$$\text{or, } a=80-b \rightarrow \text{(i)}$$

$$\text{H.M} = \frac{2ab}{a+b}$$

$$\text{or, } 30 = \frac{2ab}{a+b} \rightarrow \text{(ii)}$$

Putting the value of a in eqn (ii)

$$30 = \frac{2(80-b)b}{80-b+b}$$

$$\text{or } 30 = \frac{(160 - 2b)b}{80}$$

$$\text{or } 15 = \frac{80b - b^2}{80}$$

$$\text{or } 1200 = 80b - b^2$$

$$\text{or } b^2 - 80b + 1200 = 0$$

$$\text{or } b^2 - 60b - 20b + 1200 = 0$$

$$\text{or } b(b - 60) - 20(b - 60) = 0$$

$$\text{or } (b - 20)(b - 60) = 0$$

Either,

$$b = 20, 60$$

When $b = 20$

$$a = 80 - 20 = 60$$

When $b = 60$

$$a = 80 - 60 = 20$$

$\therefore$ Two numbers are 20 and 60 or 60 and 20.

c. The A.M between two numbers exceeds their G.M by 1 and the G.M exceeds their H.M by 0.8. Find the numbers.

→ Soln,

Let A.M and G.M be Arithmetic mean and geometric mean of two numbers a and b .

H.M - Harmonic mean of a and b .

According to question,

$$AM - GM = 1$$

$$AM = 1 + GM \rightarrow (i)$$

$$GM - H.M = 0.8$$

$$\text{or } GM = H.M + 0.8 \rightarrow (ii)$$

From (i)

$$\frac{a+b}{2} = 1 + \sqrt{ab}$$

$$\text{or } a+b = 2 + 2\sqrt{ab}$$

From (ii)

$$AM = 1 + H.M + 0.8 \quad [\because G.M = H.M + 0.8]$$

$$\text{or, } AM = H.M + 1.8 \rightarrow \text{(iii)}$$

Now,

$$GM^2 = A.M \times H.M$$

$$\text{or, } GM^2 = (1 + G.M) \times (G.M - 0.8)$$

$$\text{or, } GM^2 = GM - 0.8 + GM^2 - 0.8 G.M$$

$$\text{or, } GM^2 = 0.2 GM - 0.8 + GM^2$$

$$\text{or, } 0.2 GM = 0.8$$

$$\text{or, } G.M = \frac{0.8}{0.2}$$

$$\text{or, } G.M = 4$$

$$\text{or, } \sqrt{ab} = 4$$

Squaring on both sides,

$$ab = 16$$

$$\text{or, } a = \frac{16}{b} \rightarrow \text{(iv)}$$

From (i) and Value of G.M

$$AM = 1 + 4$$

$$\frac{a+b}{2} = 5$$

$$a+b = 10$$

$$\text{or, } a = 10 - b \rightarrow \text{(v)}$$

From (iv) and (v)

$$\frac{16}{b} = 10 - b$$

$$\text{or, } 16 = 10b - b^2$$

$$\text{or, } b^2 - 10b + 16 = 0$$

$$\text{or, } b^2 - 8b - 2b + 16 = 0$$

$$\text{or, } b(b-8) - 2(b-8) = 0$$

$$\text{or, } (b-2)(b-8) = 0$$

Either,

$$b = 2, 8$$

When $b = 2$

$$a = \frac{16}{2} = 8$$

When $b = 8$

$$a = \frac{16}{8} = 2$$

$\therefore$ Two numbers are 2 and 8 or
8 and 2

2.

Q. Find ~~the~~ ~~two~~ three numbers in A.P whose sum is 27 and product is 504.

→ Soln,

Let three numbers in A.P are

$$a-d, a, a+d$$

From question,

$$a-d + a + a+d = 27$$

$$\text{or, } 3a = 27$$

$$\text{or, } a = \frac{27}{3}$$

$$\therefore a = 9$$

Again,

$$(a-d) \times a \times (a+d) = 504$$

$$\text{or, } (9-d) \times 9 \times (9+d) = 504$$

$$\text{or, } (81-d^2) \times 9 = 504$$

$$\text{or, } 81-d^2 = \frac{504}{9} = 56$$

$$\text{or, } d^2 = 81-56$$

$$\text{or, } d^2 = 25$$

$$\text{or, } d = \sqrt{25}$$

$$\therefore d = \pm 5$$

When $d = +5$

$$a-d = 9-5 = 4$$

$$a = 9$$

$$a+d = 9+5 = 14$$

when $d = -5$

$$a-d = 9+5 = 14$$

$$a = 9$$

$$a+d = 9-5 = 4$$

$\therefore$ Three numbers are 4, 9, 14 and 14, 9, 4.

b) The Sum of three numbers in A.P is 15. When the numbers are increased by 1, 3, 9 respectively, the resulting numbers are in G.P. Find the numbers.

→ Soln,

Let 3 number in A.P be $a-d, a, a+d$

According to question,

$$a-d + a + a+d = 15$$

$$3a = 15$$

$$a = \frac{15}{3}$$

$$\therefore a = 5$$

Numbers increased by 1, 3, 9 respectively, so
 $5-d+1, 5+3, 5+9+d$ are in G.P

or, $6-d, 8, 14+d$ are in G.P

$$8 = \sqrt{(6-d)(14+d)}$$

$$\text{or, } 8 = \sqrt{84 + 6d - 14d - d^2}$$

$$\text{or, } 8 = \sqrt{84 - 8d - d^2}$$

Squaring on both side,

$$64 = 84 - 8d - d^2$$

$$\text{or, } d^2 + 8d + 64 - 84 = 0$$

$$\text{or, } d^2 + 8d - 20 = 0$$

$$\text{or, } d^2 + 10d - 2d - 20 = 0$$

$$\text{or, } d(d+10) - 2(d+10) = 0$$

$$\text{or, } (d-2)(d+10) = 0$$

Either,

$$d = -10, 2$$

When $d = -10$

$$a-d = 5+10 = 15$$

$$a = 5$$

$$a+d = 5-10 = -5$$

When $d = 2$

$$a-d = 5-2 = 3$$

$$a = 5$$

$$a+d = 5+2 = 7$$

$\therefore$ Three numbers are 3, 5, 7 or ~~15, 5, -5~~.

3.

a) If H be the H.M between a and b ,
prove that $(H-2a)(H-2b) = H^2$.

$\rightarrow$ soln,

$$H = \frac{2ab}{a+b} \Rightarrow a+b = \frac{2ab}{H} \Rightarrow ab = \frac{H(a+b)}{2}$$

Now,

$$\text{L.H.S, } (H-2a)(H-2b)$$

$$\text{or, } H^2 - 2bH - 2aH + 4ab$$

$$= H^2 - (2a+2b)H + 4ab$$

$$= H^2 - 2H(a+b) + 4ab$$

$$= H^2 - 2H \times \frac{2ab}{H} + 4ab$$

$$= H^2 - 4ab + 4ab$$

$$= H^2 = R.H.S$$

Proved

b) If H be the H.M between a and b ,
prove that $\frac{H+a}{H-a} + \frac{H+b}{H-b} = 2$

→ Soln,

$$H = \frac{2ab}{a+b}$$

$$\text{L.H.S, } \frac{H+a}{H-a} + \frac{H+b}{H-b}$$

$$= \frac{\frac{2ab}{a+b} + a}{\frac{2ab}{a+b} - a} + \frac{\frac{2ab}{a+b} + b}{\frac{2ab}{a+b} - b}$$

$$= \frac{\frac{2ab+a^2+ab}{a+b}}{\frac{2ab-a^2-ab}{a+b}} + \frac{\frac{2ab+ab+b^2}{a+b}}{\frac{2ab-ab-b^2}{a+b}}$$

$$= \frac{a^2+3ab}{ab-a^2} + \frac{b^2+3ab}{ab-b^2}$$

$$= \frac{a^2+3ab}{a(b-a)} + \frac{b^2+3ab}{-b(b-a)}$$

$$= \frac{a^2b+3ab^2 - ab^2 - 3a^2b}{+ab(b-a)}$$

$$= \frac{-2a^2b + 2ab^2}{ab(b-a)}$$

$$= \frac{2ab(b-a)}{ab(b-a)}$$

$$= 2 = R.H.S \quad \text{Proved.}$$

c) If A be the arithmetic mean and H the H.M. between a and b, prove that

$$\frac{a-A}{a-H} \times \frac{b-A}{b-H} = \frac{A}{H}$$

→ Soln,

$$A = \frac{a+b}{2}, \quad H = \frac{2ab}{a+b}$$

L.H.S,

$$\frac{a-A}{a-H} \times \frac{b-A}{b-H}$$

$$= \frac{a - \left(\frac{a+b}{2}\right)}{a - \left(\frac{2ab}{a+b}\right)} \times \frac{b - \left(\frac{a+b}{2}\right)}{b - \left(\frac{2ab}{a+b}\right)}$$

$$= \frac{\frac{2a-a-b}{2}}{\frac{a^2+ab-2ab}{a+b}} \times \frac{\frac{2b-a-b}{2}}{\frac{ab+b^2-2ab}{a+b}}$$

$$= \frac{a-b}{2} \times \frac{b-a}{2} \times \frac{a+b}{a^2-ab} \times \frac{a+b}{b^2-ab}$$

$$= \frac{a^2-b^2}{2(a^2-ab)} \times \frac{-(a^2-b^2)}{2(b^2-ab)}$$

$$= \frac{a^2 - b^2}{2a(a-b)} \times \frac{1(a^2 - b^2)}{12b(a-b)}$$

$$\Rightarrow \frac{a^2 - b^2}{2ab(a-b)}$$

$$= \frac{(a+b)(a-b)}{2a(a-b)} \times \frac{(a+b)(a-b)}{2b(a-b)}$$

$$= \frac{(a+b)^2}{4ab}$$

$$R.H.S. = \frac{A}{H}$$

$$= \frac{a+b}{2}$$

$$\frac{2ab}{a+b}$$

$$= \frac{a+b}{2} \times \frac{a+b}{2ab}$$

$$= \frac{(a+b)^2}{4ab}$$

$$\therefore L.H.S. = R.H.S.$$

d. If G is a G.M. between two positive numbers a and b , then prove that

$$\frac{1}{G^2 - a^2} + \frac{1}{G^2 - b^2} = \frac{1}{G^2}$$

→ Soln,

$$G.M. = \sqrt{ab}$$

$$L.H.S., \frac{1}{G^2 - a^2} + \frac{1}{G^2 - b^2}$$

$$= \frac{1}{(\sqrt{ab})^2 - a^2} + \frac{1}{(\sqrt{ab})^2 - b^2}$$

$$= \frac{1}{ab - a^2} + \frac{1}{ab - b^2}$$

$$= \frac{1}{a(b-a)} + \frac{1}{b(a-b)}$$

$$= \frac{1}{b(a-b)} - \frac{1}{a(a-b)}$$

$$= \frac{a-b}{ab(a-b)}$$

$$= \frac{a-b}{ab(a-b)}$$

$$= \frac{1}{ab}$$

$$= \frac{1}{(\sqrt{ab})^2}$$

$$= \frac{1}{a^2} = \text{R.H.S.} \quad \text{Proved}$$

4.

a) If a, b, c are in G.P and $a^{1/n} = b^{1/y} = c^{1/z}$ then show that n, y, z are in A.P.

→ Soln,

$$\text{let, } a^{1/n} = b^{1/y} = c^{1/z} = k$$

$$a = k^n$$

$$b = k^y$$

$$c = k^z$$

According to question,

a, b, c are in G.P

So,

$$b = \sqrt{ac}$$

$$\text{or, } K^y = \sqrt{K^n \cdot K^z}$$

$$\text{or, } K^y = \sqrt{K^{n+z}}$$

Squaring on both side,

$$K^{2y} = K^{n+z}$$

$$2y = n+z$$

$$\text{or, } y = \frac{n+z}{2}$$

$\therefore n, y, z$ are in A.P.

b) Show that b^2 is greater than, equal to, or less than ac , according as a, b, c are in A.P, G.P or H.P.

→ let a, b, c are in G.P

$$\text{i.e. } b = \sqrt{ac}$$

Squaring on both sides,

$$b^2 = ac$$

$$\therefore b^2 = ac$$

If a, b, c are in A.P then

to show $b^2 > ac$

$$b = \frac{a+c}{2}$$

Now,

$$b^2 = ac$$

$$b^2 - ac =$$

$$\Rightarrow \left(\frac{a+c}{2}\right)^2 - ac$$

$$\Rightarrow \frac{a^2 + 2ac + c^2}{4} - \frac{ac}{1}$$

$$\underline{a^2 + 2ac + c^2 - 4ac}$$

$$\text{or } \underline{a^2 - 2ac + c^2}$$

$$\text{or } \underline{(a-c)^2} \geq 0$$

$$\text{or } \underline{\left(\frac{a-c}{2}\right)^2} > 0$$

$$\text{i.e. } b^2 - ac > 0$$

$$\therefore b^2 > ac$$

proved

Next.

If a, b, c are in H.P then to show

$$b^2 < ac$$

$$b = \frac{2ac}{a+c}$$

$$ac - b^2 = ac - \left(\frac{2ac}{a+c}\right)^2$$

$$= ac - \frac{4a^2c^2}{a^2 + 2ac + c^2}$$

$$= \frac{a^3c + 2a^2c^2 + ac^3 - 4a^2c^2}{a^2 + 2ac + c^2}$$

$$= \frac{a^3c - 2a^2c^2 + ac^3}{(a+c)^2}$$

$$= \frac{ac(a^2 - 2ac + c^2)}{(a+c)^2}$$

$$= ac \left(\frac{a-c}{a+c}\right)^2 > 0$$

$$\therefore ac - b^2 > 0$$

$$\text{i.e. } ac > b^2 \Rightarrow b^2 < ac$$

$\therefore$ If a, b, c are in H.P then $b^2 < ac$

5. If a, b, c are in H.P, prove that

a) $2a-b, b, 2c-b$ are in G.P

→ Soln,

$$b^2 = (2a-b)(2c-b)$$

$$\text{or, } b^2 = 4ac - 2ab - 2bc + b^2$$

$$\text{or, } 4ac = 2ab + 2bc$$

$$\text{or, } \frac{24ac}{2abc} = \frac{2ab}{2abc} + \frac{2bc}{2abc}$$

$$\text{or, } \frac{2}{b} = \frac{1}{c} + \frac{1}{a} \quad [\text{This shows } a, b, c \text{ are in H.P}]$$

∴ This proves $2a-b, b, 2c-b$ are in G.P

b) $a(b+c), b(c+a), c(a+b)$ are in A.P

→ Soln,

∵ a, b, c are in H.P so,

$$\frac{2}{b} = \frac{1}{a} + \frac{1}{c}$$

Now,

$a(b+c), b(c+a), c(a+b)$ will be in A.P

$$\text{if, } b(c+a) = \frac{a(b+c) + c(a+b)}{2}$$

$$\Rightarrow 2bc + 2ab = ab + ac + ac + bc$$

$$\Rightarrow bc + ab = 2ac$$

$$\Rightarrow \frac{2ac}{abc} = \frac{bc}{abc} + \frac{ab}{abc}$$

$$\Rightarrow \frac{2}{b} = \frac{1}{a} + \frac{1}{c} \quad \text{Which is true.}$$

c) $\frac{a}{b+c}, \frac{b}{c+a}, \frac{c}{a+b}$ are in H.P

→ If possible,

let $\frac{a}{b+c}, \frac{b}{c+a}, \frac{c}{a+b}$ are in H.P

$$\therefore \frac{b}{c+a} = 2 \times \frac{a}{b+c} \times \frac{c}{a+b}$$

$$\frac{a}{b+c} + \frac{c}{a+b}$$

$$\text{or, } \frac{b}{c+a} = \frac{2ac}{(a+b)(b+c)}$$

$$\frac{a(a+b) + c(b+c)}{(a+b)(b+c)}$$

$$\text{or, } \frac{b}{c+a} = \frac{2ac}{a^2 + ab + bc + c^2}$$

$$\text{or, } \frac{a^2b + ab^2 + b^2c + bc^2}{a^2 + ab + bc + c^2} = \frac{2ac^2 + 2a^2c}{a^2 + ab + bc + c^2}$$

$$\text{or, } \frac{a^2b + b^2c}{a^2 + ab + bc + c^2} = \frac{2ac^2 + 2a^2c}{a^2 + ab + bc + c^2}$$

$$\text{or, } \frac{a^2b + b^2c}{a^2 + ab + bc + c^2} = \frac{2ac^2 + 2a^2c}{a^2 + ab + bc + c^2}$$

or

$$\text{or, } b = \frac{2ac(c+a)}{a^2 + ab + bc + c^2}$$

$$\text{or, } b = \frac{2ac(a+c)}{a^2 + b(a+c) + c^2}$$

$$\therefore b = \frac{2ac}{a+c}$$

$$\text{or, } b = \frac{2ac(a+c)}{a^2 + 2ac + c^2}$$

$$\Rightarrow b(a+c) = 2ac$$

$$b = \frac{2ac(a+c)}{(a+c)^2}$$

$$\therefore b = \frac{2ac}{a+c}$$

6.

a) If $\frac{1}{b+c}$, $\frac{1}{c+a}$, $\frac{1}{a+b}$ are in A.P, prove that a^2, b^2, c^2 are in A.P.

→ If possible,

let. $\frac{1}{b+c}$, $\frac{1}{c+a}$, $\frac{1}{a+b}$ are in A.P

$$\therefore \frac{1}{c+a} = \frac{\frac{1}{b+c} + \frac{1}{a+b}}{2}$$

$$\text{or, } \frac{1}{c+a} = \frac{a+b+b+c}{2(a+b)(b+c)}$$

$$\text{or, } \frac{1}{c+a} = \frac{a+2b+c}{2(ab+ac+b^2+bc)}$$

$$\text{or, } ac + \cancel{2bc} + c^2 + a^2 + \cancel{2ab} + ac = \cancel{2ab} + 2ac + 2b^2 + \cancel{2bc}$$

$$\text{or, } a^2 + \cancel{2ac} + c^2 = \cancel{2ac} + 2b^2$$

$$\text{or, } b^2 = \frac{a^2 + c^2}{2}$$

$\therefore$ This shows that $\frac{1}{b+c}$, $\frac{1}{c+a}$, $\frac{1}{a+b}$ are in A.P.

7.

a) If a, b, c be in A.P, b, c, d in G.P and c, d, e in H.P, prove that a, c, e are in G.P.

→ Soln,

a, b, c are in A.P

$$\therefore b = \frac{a+c}{2} \rightarrow (i) \quad c, d, e \text{ are in H.P.}$$

$$d = \frac{2ce}{c+e} \rightarrow (iii)$$

b, c, d are in G.P.

$$c^2 = bd \rightarrow (ii)$$

~~If possible, let a, c, e are in G.P.~~

To prove,

$$\therefore c^2 = ae$$

$$\text{or, } ~~b^2 = ac~~$$

From (ii)

$$c^2 = bd$$

$$\text{or, } c^2 = \left(\frac{a+c}{2} \right) \left(\frac{2ce}{c+e} \right)$$

$$\text{or, } c^2 = \frac{ce(a+c)}{c+e}$$

$$\text{or, } c^3 + c^3e = ace + c^3e$$

$$\text{or, } c^2 = ae$$

$$\therefore c^2 = ae$$

proved

b) If p is the A.M. between q and r ,
 q is the G.M. between r and p , then
prove that r will be the H.M. between
 p and q .

$\rightarrow$ Soln,

p is the A.M. between q and r .

$$\text{i.e. } p = \frac{q+r}{2} \rightarrow (i)$$

$$\text{or, } pq = \frac{q^2 + qr}{2}$$

q is the G.M. between r and p .

$$q^2 = rp \rightarrow (i)$$

To prove: r will be the H.M. between p and q .

$$\text{i.e. } r = \frac{2pq}{p+q}$$

From (i) and (ii)

$$q^2 = r \left(\frac{q+r}{2} \right)$$

$$\text{or, } r = \frac{q^2}{1} \times \frac{2}{(q+r)}$$

$$\text{or, } r = \frac{2q^2}{q+r}$$

$$\text{or, } qr + r^2 = 2q^2$$

$$\text{or, } qr + r^2 = 2rp$$

$$\text{or, } \cdot$$

Multiplying eqⁿ (i) by q on both side,

$$pq = \frac{q^2 + qr}{2}$$

$$\text{or, } pq = \frac{rp + qr}{2}$$

$$\text{or, } 2pq = r(p+q)$$

$$\therefore r = \frac{2pq}{p+q} \quad \text{H.P.}$$

6.b) If $\frac{1}{2}(x+y)$, y , and $\frac{1}{2}(y+z)$ be in H.P., show

that x, y, z are in G.P.

→ Soln,

$\frac{1}{2}(x+y)$, y , $\frac{1}{2}(y+z)$ are in H.P.

$$y = \frac{2 \times \frac{1}{2}(x+y) \times \frac{1}{2}(y+z)}{2}$$

$$\frac{1}{2}(x+y) + \frac{1}{2}(y+z)$$

$$\text{or, } y = \frac{(x+y)(y+z)}{2}$$

$$\frac{x+y+z}{2}$$

$$\text{or, } y = \frac{(x+y)(y+z)}{x+y+z}$$

$$\text{or, } y = \frac{xy + xz + y^2 + yz}{x+y+z}$$

$$\text{or, } \cancel{xy} + 2y^2 + \cancel{yz} = \cancel{xy} + xz + y^2 + yz$$

$$\text{or, } 2y^2 - y^2 = xz$$

$$\text{or, } y^2 = xz$$

$$\text{or, } y = \sqrt{xz}$$

∴ This shows that x, y, z are in G.P.

c) If $x+y, 2y, y+z$ are in H.P., prove that x, y, z are in G.P.

→ Soln,

$x+y, 2y, y+z$ are in H.P.

$$\text{i.e. } 2y = \frac{2(x+y)(y+z)}{x+y+y+z}$$

$$\text{or, } y = \frac{xy + xz + y^2 + yz}{x+2y+z}$$

$$\text{or, } \cancel{xy} + 2y^2 + \cancel{yz} = \cancel{xy} + xz + y^2 + \cancel{yz}$$

$$\text{or, } 2y^2 - y^2 = xz$$

$$\text{or, } y^2 = xz$$

∴ This shows that x, y, z are in G.P.

d) If a, b, c are in G.P., a, x, b are in A.P. and b, y, c are in A.P. prove that $\frac{1}{x} + \frac{1}{y} = \frac{2}{b}$

→ Soln,

a, b, c are in G.P.

$$\text{i.e. } b^2 = ac$$

a, x, b are in A.P.

$$\text{i.e. } x = \frac{a+b}{2}$$

b, y, c are in A.P.

$$\text{i.e. } y = \frac{b+c}{2}$$

$$\text{L.H.S.}, \frac{1}{x} + \frac{1}{y}$$

$$= \frac{1}{\frac{a+b}{2}} + \frac{1}{\frac{b+c}{2}}$$

$$= \frac{2}{a+b} + \frac{2}{b+c}$$

$$= \frac{2b+2c+2a+2b}{(a+b)(b+c)}$$

$$= \frac{2a+4b+2c}{(a+b)(b+c)}$$

$$= \frac{2a+4b+2c}{(a+b)(b+c)}$$

$$= \frac{b(2a+4b+2c)}{b(ab+ac+b^2+bc)}$$

$$= \frac{2b(2b+a+c)}{b(ab+ac+ac+bc)} \quad [\because b^2 = ac]$$

$$= \frac{2b(a+2b+c)}{b(ab+2ac+bc)}$$

$$= \frac{2b(a+2b+c)}{b(ab+2ac+bc)}$$

$$= \frac{2b(a+2b+c)}{ab^2+2abc+b^2c}$$

$$= \frac{2b(a+2b+c)}{ab^2+2abc+b^2c}$$

$$= \frac{2b(a+2b+c)}{a^2c+2abc+ac^2} \quad [\because b^2 = ac]$$

$$= \frac{2b(a+2b+c)}{a^2c+2abc+ac^2}$$

$$= \frac{2b(a+2b+c)}{ac(a+2b+c)}$$

$$= \frac{2b}{b^2} \quad [\because b^2 = ac]$$

$$= \frac{2}{b}$$

$$= \frac{2}{b}$$

∴

7.c) If p^{th} , q^{th} , r^{th} terms of an A.P are a, b, c respectively, then show that.

$$a(q-r) + b(r-p) + c(p-q) = 0$$

→ Given,

$$t_p = a$$

$$t_q = b$$

$$(t_r = c)$$

$$\text{L.H.S} = a(q-r) + b(r-p) + c(p-q) = 0$$

$$= [A + (p-1)d](q-r) + [A + (q-1)d](r-p) + [A + (r-1)d](p-q)$$

$$= A(q-r) + (p-1)(q-r)d + A(r-p) + (q-1)(r-p)d + A(p-q) + (r-1)(p-q)d$$

$$= A(q-r) + (p-1)(q-r)d + A(r-p) + (q-1)(r-p)d + A(p-q) + (r-1)(p-q)d$$

$$= A(q-r) + (p-1)(q-r)d + A(r-p) + (q-1)(r-p)d + A(p-q) + (r-1)(p-q)d$$

$$= A(q-r) + (p-1)(q-r)d + A(r-p) + (q-1)(r-p)d + A(p-q) + (r-1)(p-q)d$$

$$= 0 + d(pq - q - pr + r) + d(qr - r - pq + p) + d(pr - qr - p + q)$$

$$= 0 + d(pq - q - pr + r) + d(qr - r - pq + p) + d(pr - qr - p + q)$$

$$= 0 + d(pq - q - pr + r) + d(qr - r - pq + p) + d(pr - qr - p + q)$$

$$= (pq - q - pr + r + qr - r - pq + p + pr - qr - p + q) d$$

$$= 0 \times d$$

$$= 0 = \text{R.H.S}$$

proved

8. If three unequal positive numbers a, b, c are in H.P. prove that.

$$a) a^2 + c^2 > 2b^2$$

→ Given,

a, b, c are in H.P then,

$$b^2 < ac \quad \text{i.e. } ac > b^2$$

$$\text{To show: } a^2 + c^2 > 2b^2$$

we have,

$$(A.M \text{ between } a^2 \text{ \& } c^2) > (G.M \text{ between } a^2 \text{ \& } c^2)$$

$$\text{i.e. } \frac{a^2 + c^2}{2} > \sqrt{a^2 c^2}$$

$$\text{i.e. } a^2 + c^2 > 2ac > 2b^2 \quad (\because ac > b^2)$$

$$\therefore a^2 + c^2 > 2b^2$$

Proved

$$b) \quad a^3 + c^3 > 2b^3$$

→ If a, b, c are in H.P. then

$$b^2 < ac \quad \text{i.e. } ac > b^2$$

$$\text{To show: } a^3 + c^3 > 2b^3$$

we have

$$A.M \text{ between } a^3 \text{ \& } c^3 > G.M \text{ between } a^3 \text{ \& } c^3$$

$$a^3 \text{ \& } c^3$$

$$\text{i.e. } \frac{a^3 + c^3}{2} > \sqrt{a^3 c^3}$$

$$\text{i.e. } a^3 + c^3 > 2\sqrt{a^3 c^3}$$

$$\text{i.e. } a^3 + c^3 > 2(ac)^{3/2} > 2(b^2)^{3/2}$$

$$a^3 + c^3 > 2b^3$$

$$\therefore a^3 + c^3 > 2b^3$$

Proved

$$c) a^n + c^n > 2b^n$$

↳ If a, b, c are in H.P then,

$$b^2 < ac \text{ i.e. } ac > b^2$$

To show: $a^n + c^n > 2b^n$

We have.

A.M between a^n & c^n > G.M between a^n & c^n

$$\text{i.e. } \frac{a^n + c^n}{2} > \sqrt{a^n c^n}$$

$$\text{or, } a^n + c^n > 2(ac)^{n/2} > 2(b^2)^{n/2}$$

$$a^n + c^n > 2b^n$$

$$\therefore a^n + c^n > 2b^n$$

S. If a, b, c are in H.P, prove that